

ARVINDER S DHOUL

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scholar.google.com/citations?user=ia-DnV0AAAAAJ | arvinder-portfolio-smoky.vercel.app

Summary

MSc AI & Machine Learning student at the University of Limerick and AWS Certified Solutions Architect. I enjoy taking projects from the drawing board to production, with hands-on experience building cloud backend systems, Generative AI tools, and agentic workflows. I care deeply about writing clean, maintainable code and working closely with teams to turn complex problems into simple, user-focused solutions. I am also a published IEEE researcher with papers in Computer Vision and Federated Learning.

Education

University of Limerick <i>Master of Science in Artificial Intelligence & Machine Learning</i>	Sept. 2026 – Aug. 2027 (Expected) <i>Limerick, Ireland</i>
Gyan Ganga Institute of Technology and Sciences <i>Affiliated to RGPV, Bhopal</i> <i>B.Tech in Computer Science & Business Systems</i> 8.24 CGPA · First Division with Honours	2022 – 2026 <i>Jabalpur, India</i>

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C/C++
AI & Machine Learning: PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph
Backend & Data: FastAPI, Next.js, WebSockets, MongoDB
Cloud & DevOps: AWS, Docker, Linux, CI/CD
Methodologies: Agile/Scrum, Object-Oriented Design, System Design

Certifications

AWS Certified Solutions Architect – Associate Amazon Web Services (AWS) Credential ID: 2efb1c6b2342424dbc3620b3341af42b	May 2026 – May 2029
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Experience

Compucom CSI Systems <i>AI Engineer Intern</i> - Built real-time LLM voice pipelines and RAG workflows, owning features from strategy to deployment. - Built resilient REST and WebSocket backends with Redis, encrypting PII to ensure GDPR compliance. - Wrote clean code in an agile team and managed GitHub Actions CI/CD to ensure operational excellence.	Feb 2026 – Sept 2026 <i>Remote</i>
RIU Global <i>Freelance Web Developer</i> <i>riuglobal.com</i> - Built a Next.js and MongoDB app for an international client, ensuring zero downtime at launch. - Created a custom dashboard by building robust API integrations and scalable software services to independently manage course content. - Managed cloud hosting, DNS, and database indexing to optimize delivery and monitor production.	Nov 2025 – Jan 2026 <i>Remote</i>

Projects

AgenticRAG <i>LangGraph, LangChain, Qdrant, Next.js, TypeScript, MongoDB</i> - Engineered a LangGraph agent that evaluates relevance and re-queries for context accuracy. - Mapped generated answers to exact source chunks and passages, eliminating AI hallucination. - Built a Next.js interface with Qdrant indexing, inline citations, and document previews.	GitHub Live
HireIQ <i>Python, FastAPI, TypeScript, React, MongoDB, Redis, Docker</i> - Built a scalable FastAPI & React platform using Groq/Gemini LLMs to dynamically generate and score interviews. - Engineered real-time HR dashboards via Redis Pub/Sub and Server-Sent Events (SSE) for sub-second state synchronization. - Integrated asynchronous MongoDB and a custom SMTP pipeline, deploying the full-stack architecture to Render and Vercel.	GitHub Live
Customer Churn Risk Dashboard <i>XGBoost, Scikit-Learn, Pandas, Python</i> - Trained an XGBoost model on 100000+ records, applying structured thinking to beat baselines. - Built a dashboard enabling stakeholders to test strategies and view actionable insights. - Used feature importance and cross-validation to identify churn drivers and break down problems.	GitHub Live

Research

1. Evaluating YOLOv8, MobileNet-SSD, and MTCNN for Face Detection in Smart Attendance Systems <i>Published</i>	IEEE Xplore
2. Generative AI-Powered COBOTs for Human-Centric Collaboration and Co-Evolution <i>In Publication</i>	Scopus
3. Privacy-Preserving Naval Ship Detection and Authorization Using YOLOv8 and Federated Learning	in Preparation
4. A Hybrid YOLOv8-CNN-LBPH Framework for Real-Time Intrusion and Weapon Detection	in Preparation

Extracurricular

- GATE 2025 Qualified** (Computer Science & IT) and **NPTEL Elite Certified** in Database Management Systems and Ethical Hacking.
- Hackathon Finalist:** Finalist at Smart India Hackathon 2023 and Solve to Evolve Hackathon at IIT Madras.
- Open Source & Problem Solving:** Actively maintain open-source GitHub projects and practice data structures and algorithms (DSA) problems.